

CSS Advanced

Week 3 · Friday Session

→ CSS Position

→ Pseudo-classes & Pseudo-elements

→ Transitions & Animations

→ Custom Properties

→ Media Queries

→ CSS Grid Introduction

Today's Agenda

WEEK 3 · FRIDAY

What we cover in this Friday 2-hour session

CSS Position

static · relative · absolute · fixed · sticky

CSS Custom Properties

--variables · var() · theming with :root

Pseudo-classes & Elements

:hover · :focus · :nth-child · ::before · ::after

Media Queries

@media · responsive breakpoints · mobile-first

Transitions & Animations

transition · @keyframes · animation shorthand

CSS Grid Introduction

display:grid · template-columns · gap

CSS Position

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Five values that control how an element is placed in the layout

position:
static

Default — normal flow
No
top/left/right/bottom
effect

position:
relative

Normal flow THEN
offset
from its own original
spot

position:
absolute

Removed from flow
Anchored to nearest
positioned parent

position:
fixed

Removed from flow
Stays fixed to the
VIEWPORT

position:
sticky

Relative until scroll
threshold → then fixed

CSS

```
/* position affects HOW coordinates apply */  
.elem {  
  position: absolute; /* removes from flow */  
  top:    20px;  
  right:  20px;  
}
```

absolute vs fixed — The Key Difference

One anchors to a parent, the other to the browser window

position: absolute

- Removed from normal flow
- Positioned relative to nearest POSITIONED ancestor (parent with position ≠ static)
- If no positioned ancestor → positions relative to <body>
- SCROLLS with the page

CSS

```
.parent { position: relative; }  
.tooltip { position: absolute;  
  top: 0; right: 0; }
```

position: fixed

- Removed from normal flow
- Positioned relative to the VIEWPORT (browser window)
- Does NOT move when the page scrolls
- Perfect for sticky navbars, chat buttons, cookie banners

CSS

```
nav {  
  position: fixed;  
  top: 0; left: 0;  
  width: 100%;  
}
```

position: sticky — The Smart One

Behaves as relative until a threshold → then acts like fixed

`position: sticky` is a HYBRID. Before the element reaches the specified offset during scrolling it sits in the normal flow (like relative). Once the scroll position hits the threshold (e.g., `top: 0`), it sticks in place (like fixed) — but ONLY within its parent container. When you scroll past the parent, the element unsticks.

CSS

```
/* Section heading that sticks as you scroll */
.section-title {
  position: sticky;
  top: 60px; /* sticks 60px from the top of the viewport */
  background: #0f766e;
  z-index: 10;
}
```

Common uses

- Table headers that stay visible while scrolling rows
- Section labels in long scrollable lists
- Sidebar that sticks until the section ends
- Progress bar at top that appears after scroll

Check-In · CSS Position

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Q1–Q5 · Answer each before reading the next one

Q1 What is the DEFAULT value of the CSS position property?

- ☐ static
- ☐ relative
- ☐ absolute
- ☐ fixed

Q2 position: relative offsets the element relative to ...

- ☐ its nearest positioned ancestor
- ☐ the viewport
- ☐ its own original position
- ☐ the document <body>

Q3 position: absolute anchors the element relative to ...

- ☐ the viewport
- ☐ its nearest POSITIONED ancestor (or <body>)
- ☐ the document root
- ☐ the screen

Q4 Which position value stays fixed to the viewport while scrolling?

- ☐ absolute
- ☐ relative
- ☐ fixed
- ☐ sticky

Q5 position: sticky switches from relative to fixed when ...

- ☐ the element is clicked
- ☐ z-index becomes positive
- ☐ the scroll reaches the element's threshold
- ☐ JavaScript sets it

Check-In · CSS Position

WEEK 3 · FRIDAY

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Q1 What is the DEFAULT value of the CSS position property?

✓ **static**

relative
absolute
fixed

Q2 position: relative offsets the element relative to ...

its nearest positioned ancestor
the viewport

✗ **its own original position**

the document <body>

Q3 position: absolute anchors the element relative to ...

the viewport

✗ **its nearest POSITIONED ancestor (or <body>)**

the document root
the screen

Q4 Which position value stays fixed to the viewport while scrolling?

absolute
relative

✗ **fixed**

sticky

Q5 position: sticky switches from relative to fixed when ...

the element is clicked
z-index becomes positive

✗ **the scroll reaches the element's threshold**

JavaScript sets it

CSS Pseudo-classes

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Style an element based on its STATE — no JavaScript needed

:hover

User hovers the mouse over the element

```
a:hover { color: teal; }
```

:focus

Element receives keyboard/click focus

```
input:focus { outline: 2px solid teal; }
```

:visited

Link has already been visited

```
a:visited { color: purple; }
```

:active

Element is being clicked RIGHT NOW

```
button:active { transform: scale(0.97); }
```

:first-child

First child of its parent

```
li:first-child { font-weight: bold; }
```

:nth-child(n)

Every nth child ($2n = \text{even}$, $2n+1 = \text{odd}$)

```
tr:nth-child(even) { background: #f0fdfa; }
```

CSS Pseudo-elements

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::before & ::after — insert virtual elements with CSS alone

::before — content BEFORE the element's text

- Inserts a virtual element BEFORE the content
- Must have content: property to render
- Can be styled just like a real element
- No extra HTML needed

CSS

```
h2::before {  
  content: ' 🌿 ';  
  color: #14b8a6;  
}  
/* Renders: 🌿 My Heading */
```

::after — content AFTER the element's text

- Inserts a virtual element AFTER the content
- Same rules: must have content: property
- Popular for decorative lines, badges, icons
- Used in CSS-only tooltip patterns

CSS

```
.required::after {  
  content: ' *';  
  color: red;  
  font-size: 0.9em;  
}
```

Check-In · Pseudo-classes, z-index, overflow

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Q6–Q12

Q6 z-index controls ...

stacking order (which element is on top)
element zoom
opacity
3D depth in perspective

Q7 overflow: hidden on a container ...

clips content beyond the container bounds
hides the element completely
makes content scrollable
collapses the height to zero

Q8 Which pseudo-class fires when a mouse hovers?

:hover
:focus
:active
:visited

Q9 :focus applies when ...

the element receives keyboard or click focus
the mouse hovers
a link was already visited
the element is the first child

Q10 :nth-child(2n) selects ...

every even-numbered child
every 2nd type element
children after the 2nd
only the 2nd child

Q11 ::before inserts content ...

before the element's own text/content
before the parent element
as the last child
as a sibling after the element

Q12 ::before and ::after MUST always have which property to be visible?

✓ content: (even if empty string "")
display: block
position: absolute
z-index: 1

Check-In · Pseudo-classes, z-index, overflow

WEEK 3 · FRIDAY

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position: absolute
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CSS Transitions

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Smoothly animate a CSS property change over time

transition: property duration timing-function delay;

property

WHAT to animate
(background-color, transform,
opacity ...)

duration

HOW LONG (0.3s, 500ms ...)

timing

EASING CURVE
(ease · linear · ease-in · ease-out)

delay

WAIT before starting
(0s, 200ms ...)

CSS

```
/* Smooth hover button – the classic use */
.btn {
  background: #0f766e;
  color: white;
  transition: background 0.25s ease, transform 0.15s ease;
}
.btn:hover {
  background: #14b8a6; /* transition animates this change */
  transform: translateY(-2px);
}
```

Best practice

→ Use 0.2s–0.4s for UI transitions
(fast enough to feel instant, slow enough to see)

→ `transition: all` works but is slow
— name the specific property

→ You can list multiple transitions
separated by commas

CSS Animations — @keyframes

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Multi-step animations that run automatically, no JavaScript needed

transition = A → B (triggered by state change) animation = A → B → C → ... (runs on its own)

CSS

```
/* STEP 1: define the keyframe sequence */
@keyframes fadeIn {
  from { opacity: 0; transform: translateY(20px); }
  to   { opacity: 1; transform: translateY(0); }
}

/* STEP 2: attach to the element */
.hero {
  animation: fadeIn 0.6s ease-out forwards;
}

/* Multiple stops: 0% 33% 66% 100% */
@keyframes pulse {
  0%, 100% { transform: scale(1); }
  50%     { transform: scale(1.05); }
}
```

animation shorthand parts

name	fadeIn — must match @keyframes name
duration	0.6s — how long one cycle takes
timing	ease-out · linear · cubic-bezier(...)
delay	0s — wait before starting
iteration	1 · infinite · 3
direction	normal · reverse · alternate
fill-mode	forwards (keep final state) none · backwards · both

Check-In · Transitions, Animations, Variables

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Q13–Q17

Q13 The CSS transition property animates changes in ...

- CSS property values when they change
- HTML element positions
- JavaScript variable values
- DOM structure

Q14 Which at-rule defines a CSS animation's keyframe sequence?

- @keyframes
- @media
- @font-face
- @animate

Q15 CSS custom properties (variables) are declared with the syntax ...

- name: value;
- \$name: value;
- var(name): value;
- name = value;

Q16 To USE a CSS variable called --primary, the syntax is ...

- var(--primary)
- \$(--primary)
- #primary
- %(primary)

Q17 @media (max-width: 768px) { } applies styles when ...

- the viewport is 768px wide or NARROWER
- the viewport is exactly 768px
- the font is 768px
- the device is mobile-only

Check-In · Transitions, Animations, Variables

WEEK 3 · FRIDAY

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- the font is 768px
- the device is mobile-only

CSS Custom Properties (Variables)

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Define once in :root, use everywhere with var()

CSS

```
/* 1. DECLARE variables on :root so they're globally available */
:root {
  --primary:    #0f766e;    /* teal */
  --accent:     #14b8a6;
  --bg:         #f0fdfa;
  --text:       #1e293b;
  --radius:     8px;
  --shadow:     0 4px 16px rgba(0,0,0,0.12);
}

/* 2. USE with var() */
.btn { background: var(--primary); border-radius: var(--radius); }
.card { background: var(--bg); box-shadow: var(--shadow); }
h1 { color: var(--primary); }

/* 3. DARK MODE: just redefine the same variables */
@media (prefers-color-scheme: dark) {
  :root {
    --bg:    #051a0a;
    --text:  #d4edda;
  }
}
```

Why variables?

One change updates everywhere

Change --primary once and every button, link, and card updates instantly.

Easy dark mode

Redefine variables in a @media query: no class changes, no JavaScript needed.

Component theming

Pass a different value per component:
.card { --accent: red; }

Runtime access via JS

```
document.documentElement.style
.setProperty('--primary', 'blue')
```

CSS Media Queries

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Apply styles only when certain conditions are met (responsive design)

```
@media ( condition ) { /* CSS rules that only apply when condition is true */ }
```

Mobile

```
@media (max-width: 480px)
```

≤ 480px

Tablet

```
@media (max-width: 768px)
```

≤ 768px

Desktop

```
@media (min-width: 1024px)
```

≥ 1024px

Wide

```
@media (min-width: 1440px)
```

≥ 1440px

CSS

```
/* Mobile-first approach – START with mobile, ADD complexity */
/* Base styles target mobile */
.grid { display: block; }

@media (min-width: 768px) {
  .grid { display: flex; gap: 20px; } /* tablet: row */
}

@media (min-width: 1024px) {
  .grid { display: grid; grid-template-columns: repeat(3, 1fr); }
}
```

Other media features

- (orientation: landscape)
- (prefers-color-scheme: dark)
- (prefers-reduced-motion)
- print – styles for printing
- screen – default, any display

Flexbox — Deeper Dive

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flex-direction · flex-wrap · order · flex-grow · gap

flex-direction

row (default) / column
row-reverse / column-reverse

flex-wrap

nowrap (default) / wrap
wrap-reverse

order

Default: 0 · lower = earlier
Visual reorder without changing HTML

flex-grow

0 (don't grow) or 1 (fill space)
flex: 1 = grow:1, shrink:1, basis:0

align-self

Override align-items for ONE item
auto · flex-start · center · flex-end

gap

Space between items (row & column)
Replaces margin hacks — cleaner

CSS

```
.container { display:flex; flex-wrap:wrap; gap:16px; }  
.item      { flex: 1; min-width: 200px; }  
.featured { order: -1; flex: 2; } /* appears first, twice as wide */
```

CSS Grid — Introduction

WEEK 3 · FRIDAY

Two-dimensional layout: rows AND columns at the same time

Flexbox = 1 axis (row OR column) Grid = 2 axes (rows AND columns together)

CSS

```
.grid {
  display: grid;
  grid-template-columns: repeat(3, 1fr); /* 3 equal cols */
  grid-template-rows: auto auto;        /* rows = content height */
  gap: 20px;                             /* both row & col gap */
}

/* Span multiple cells */
.hero { grid-column: 1 / 3; } /* spans cols 1 to 3 */
.sidebar { grid-row: 1 / 3; } /* spans rows 1 to 3 */

/* Named areas — even cleaner */
.layout {
  grid-template-areas:
    'header header header'
    'sidebar main main'
    'footer footer footer';
}
.main-content { grid-area: main; }
```

Key properties

grid-template-columns

Defines column widths
repeat(3,1fr) · 200px 1fr 200px

grid-template-rows

Defines row heights
auto · 100px · minmax(100px,auto)

gap (or column-gap)

Space between cells
16px or 12px 20px (row col)

grid-column

Which columns item spans
1 / 3 or span 2

grid-row

Which rows item spans
1 / 4 or span 2

grid-template-areas

Named layout zones — most
readable way to define a page layout

Check-In · Grid, Flexbox Advanced, border-radius, box-shadow

WEEK 3 · FRIDAY

Q18 flex-direction: column makes flex items ...

stack vertically (a column)
wrap to the next row
reverse their order
shrink to minimum width

Q19 flex-wrap: wrap allows flex items to ...

move to a new row when they overflow
shrink below minimum size
overlap each other
ignore the gap property

Q20 display: grid on a container ...

activates the 2-dimensional Grid layout
makes children flex items
hides overflowing children
stacks children absolutely

Q21 grid-template-columns: repeat(3, 1fr) creates ...

3 equal-width columns
1 column of 3fr width
3 rows
a fixed 300px grid

Q22 Which property adds space between grid OR flex items?

gap
margin
padding
spacing

Q23 border-radius: 50% on a square element creates ...

a perfect circle
a rounded rectangle
an oval
a hexagon

Q24 In box-shadow: 2px 4px 8px rgba(0,0,0,0.2) the 3rd value (8px) controls ...

blur radius
horizontal offset
vertical offset
spread radius

Q25 The CSS inherit keyword makes a property ...

take the computed value from the parent element
reset to browser default
apply !important override
copy from the next sibling

Check-In · Grid, Flexbox Advanced, border-radius, box-shadow

WEEK 3 · FRIDAY

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3 rows
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vertical offset
spread radius

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reset to browser default
apply !important override
copy from the next sibling

What you can now do:

Position

Use static/relative/absolute/fixed/sticky to place elements precisely

Pseudo

Style elements by state (:hover :focus) and insert content (::before ::after)

Transition

Animate property changes smoothly with transition: property duration easing

Animation

Create multi-step effects with @keyframes and the animation shorthand

Variables

Define --custom-properties in :root and use them everywhere with var()

Responsive

Use @media (max-width: ...) to apply styles only at certain viewport sizes

Grid

Create 2-dimensional layouts with display:grid and grid-template-columns